Project Seminar WS2026/27





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 4: Analog circuits and project work

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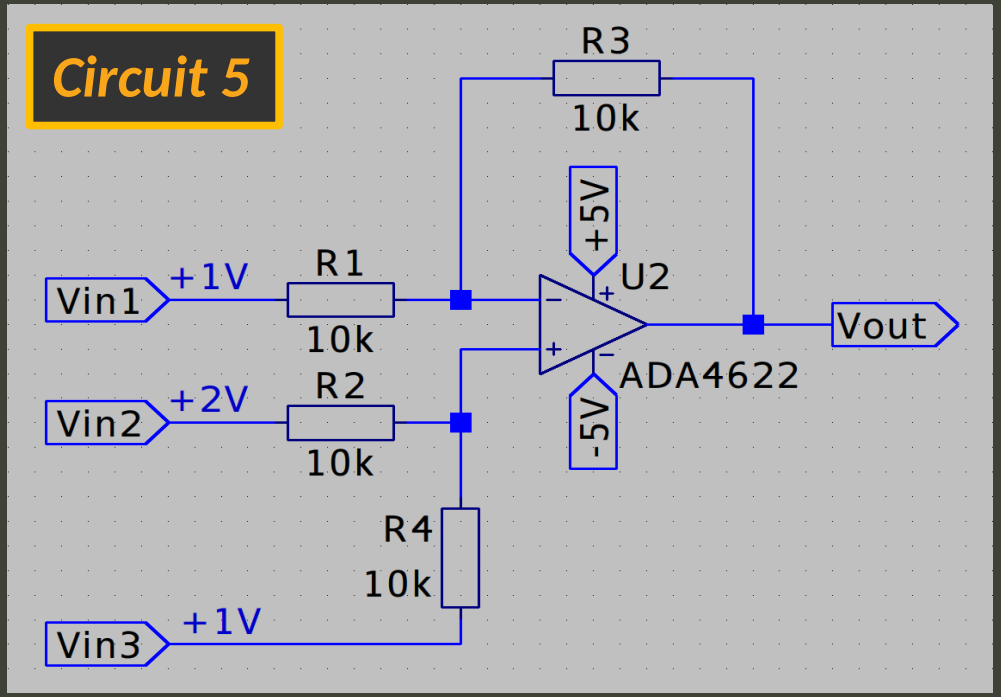
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- ***Exercise 8: → Differential operational amplifier:***
  - ADA4622 as differential operational amplifier with reference voltage

→ Task: How can you calculate the output voltage Vout? Figure it out!



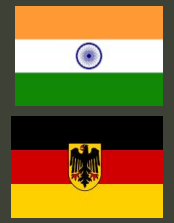
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- ***Analog circuits → exercises:***
  - Exercise 9: What is a Hall sensor? How can a Hall sensor be used to measure the current flowing through a cylindrical copper conductor? Provide a comprehensive written explanation of the measurement principle, including the relevant formulas for calculating the Hall voltage (output voltage of the hall-sensor)!

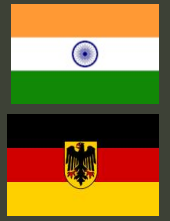
→ TASK: Answer these questions in your report!

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- ***Analog circuits & PCB-design → exercises:***
  - Exercise 10: Study the datasheets for the following components and summarize in writing all key information needed to understand how they work, as well as to create the components symbols and the corresponding footprints in KiCad ! :
    → 1. WP-THRBU REDCUBE 74650073
    → 2. RT026xxHBLU Typ 073 (→ “AKL073-2“)
    → 3. mpe garry 087 pinhead connector
    → 4. BCS-122-L-S-TE
    → 5. ACS37010LLZATR-030U5
    → 6. MAX49925XATB+T
    → 7. solder pin “RTM_1_3” (from company Vogt article number 1365.6)

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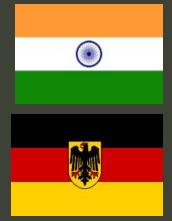


- ***Analog circuits & PCB-design → Exercise 10 continued:***

  → 8. AD8031ARZ
  → 9. AD8032ARZ
  → 10. MAX6070AAUT41+T
  → 11. MAX6070AAUT21+T
  → 12. ADS8329IPWR
  → 13. ADA4099-1BUJZ-RL7
  → 14. WPME-CDIP 18024115401L
  → 15. WPME-CDIP 18024215401L
  → 16. ESP32-S3-DevkitC-1N8R8

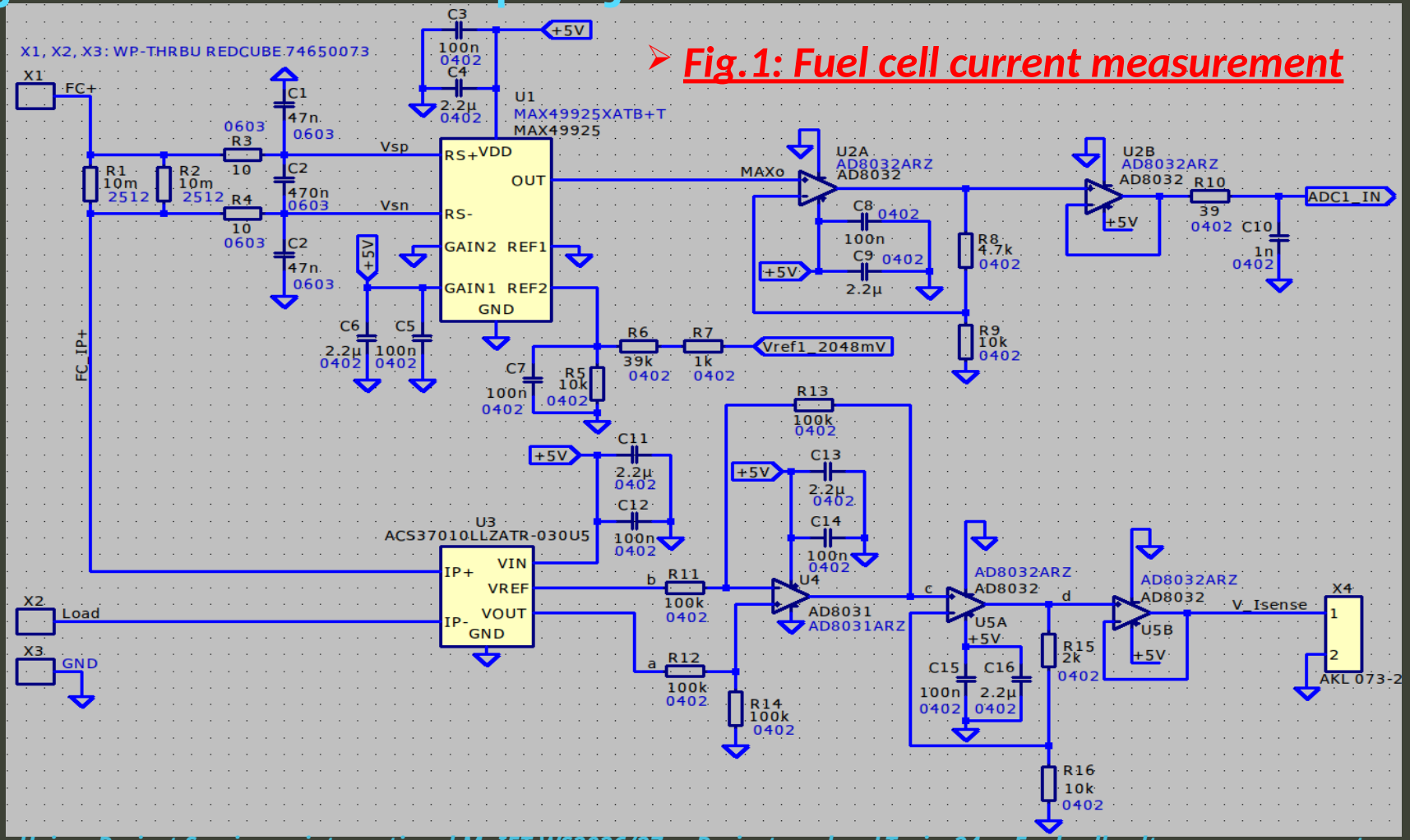
→ TASK: Put your solutions in your report!

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- ***Analog circuits → Overview:***
  - The circuits which are shown on the following slides are related to the project and are sections from the whole simulation schematic created in LTspice24.
  - The simulation of the fuel cell monitoring and measurement board in LTspice24 is not a part of the project seminar.
  - One of the goals in this project seminar is to transfer the schematic sections from LTspice24 into a schematic in KiCad.

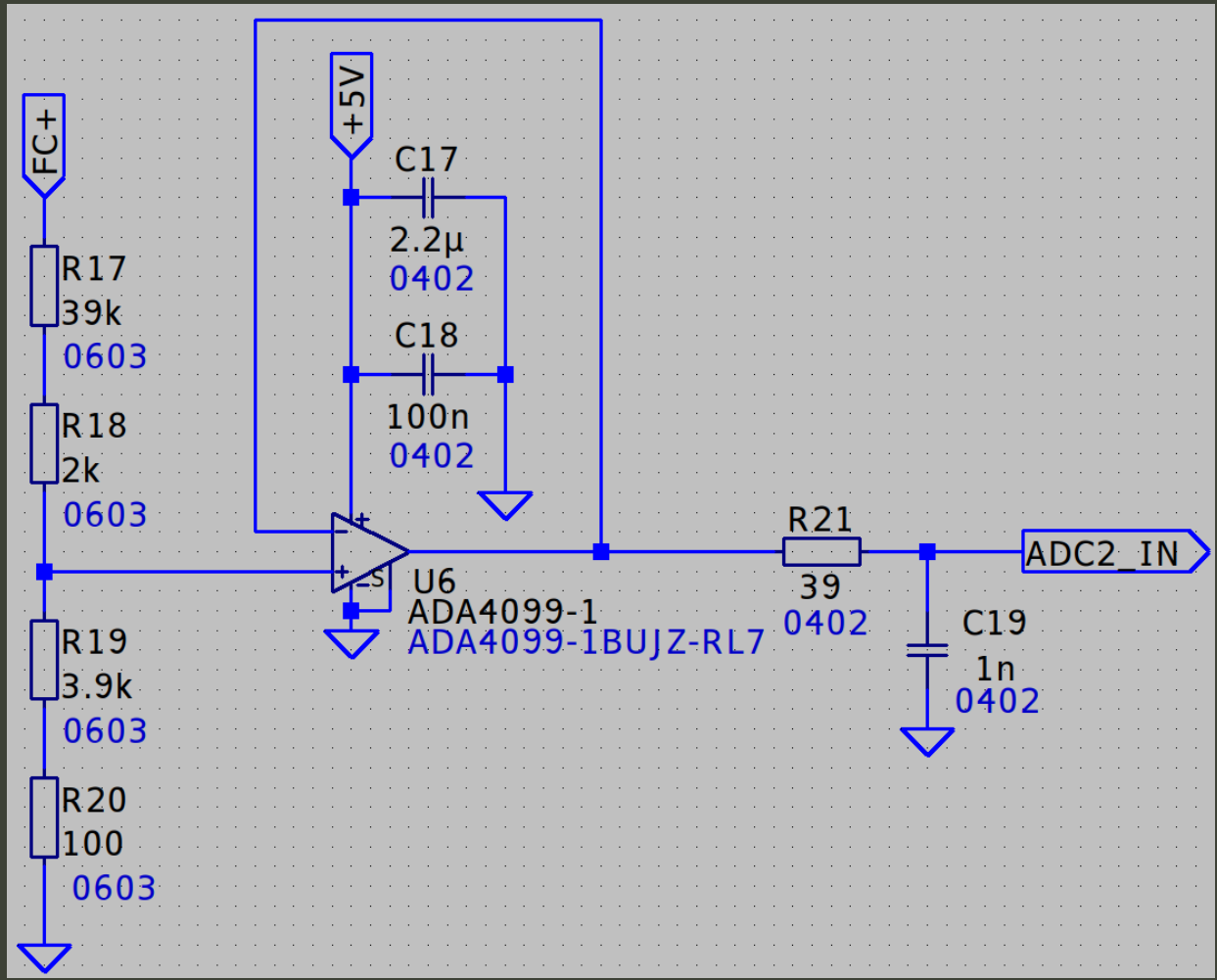
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➢ ***Fig.1: Fuel cell current measurement***

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- ***Fig. 2:***
  Fuel cell voltage measurement

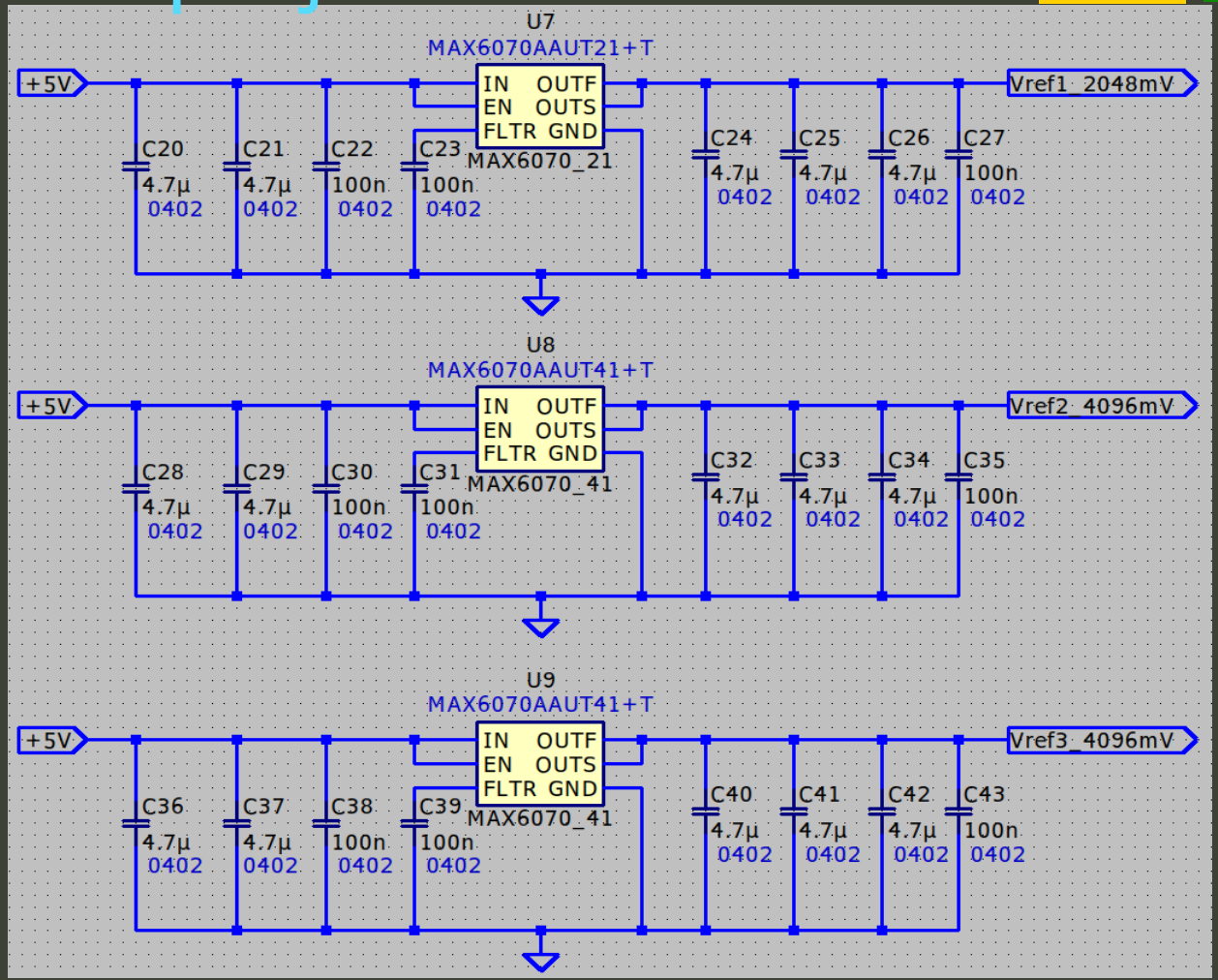


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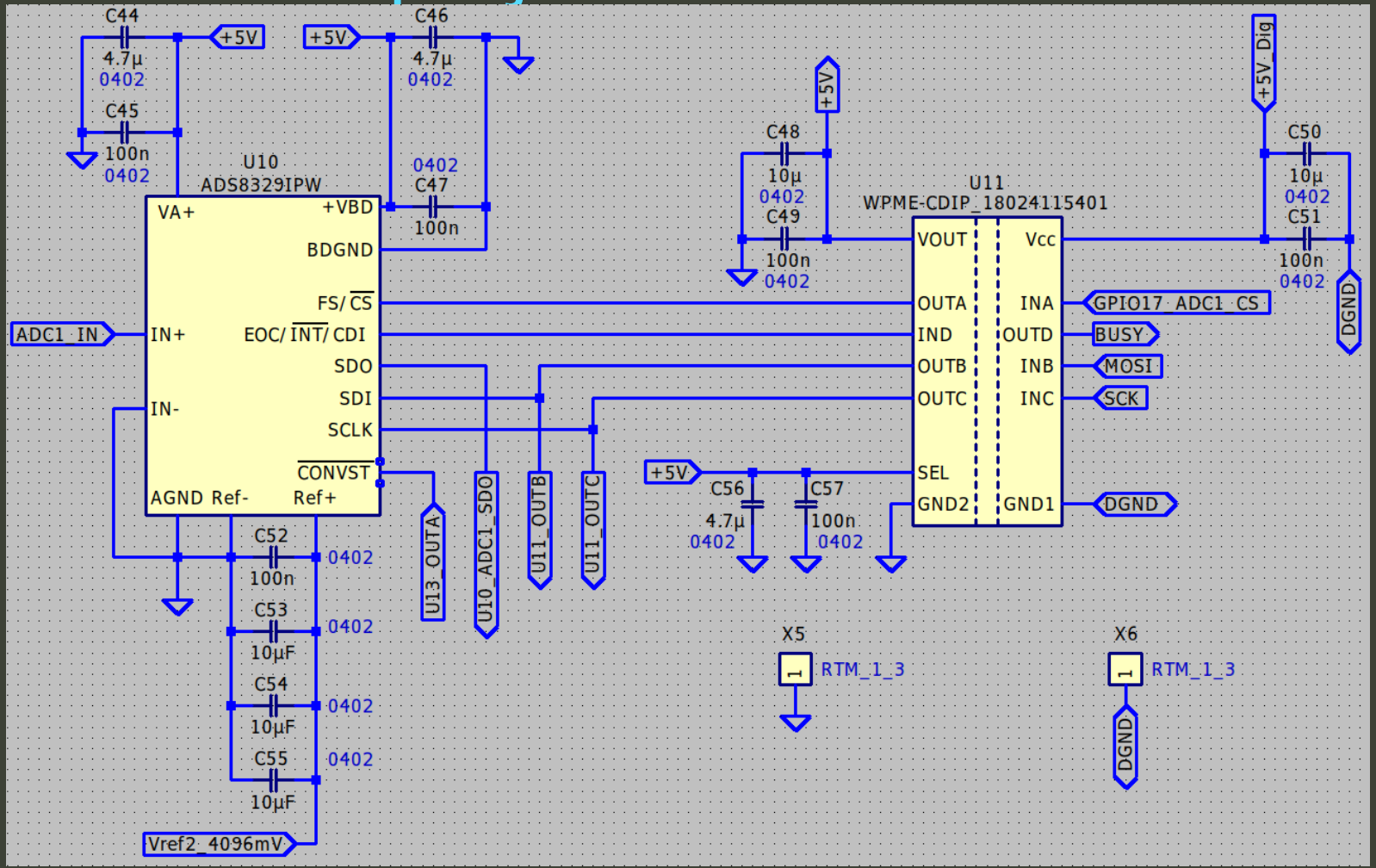


- ***Fig.3:***
  Reference voltages



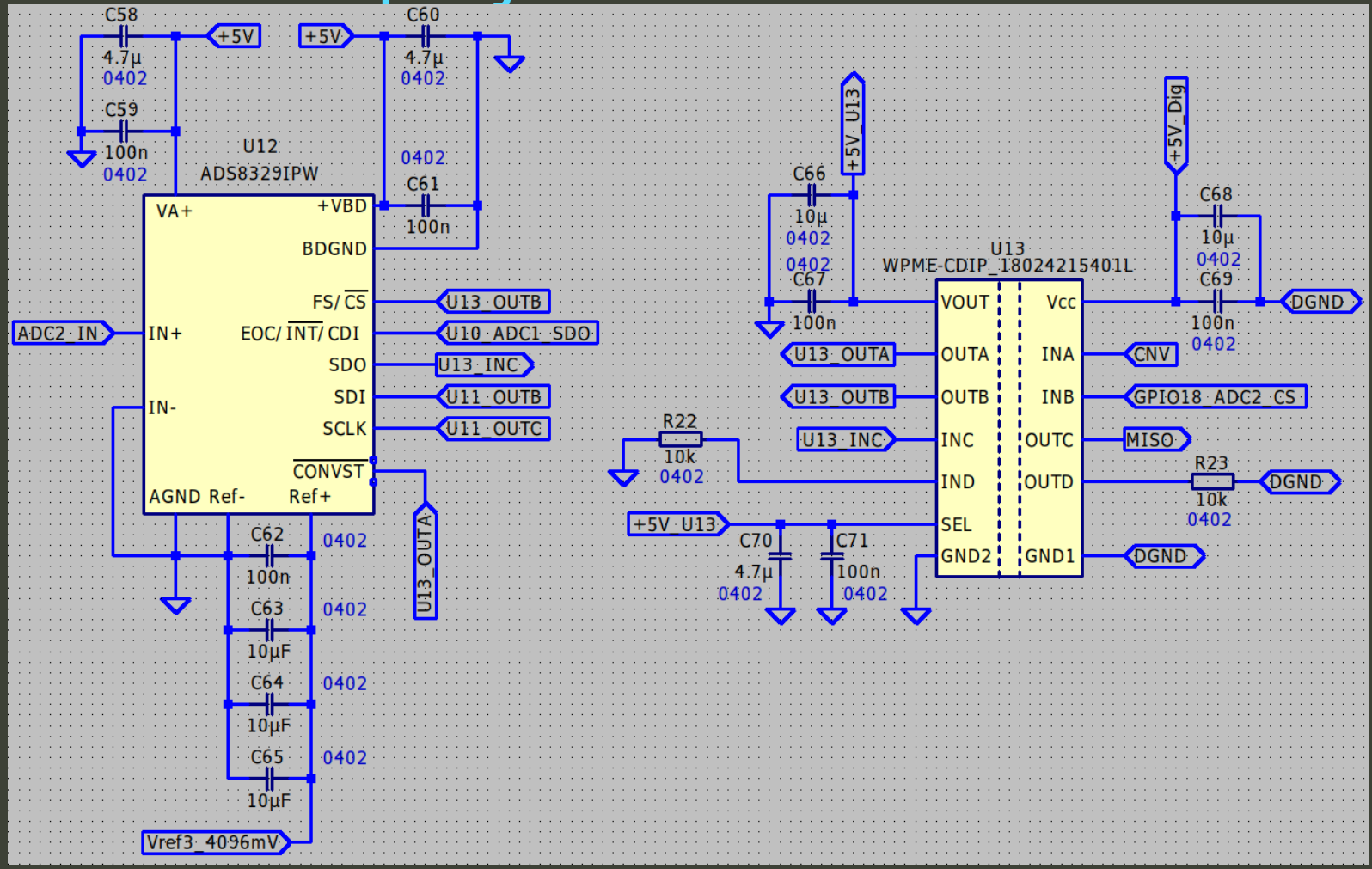
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- ***Fig. 4: ADC #1***



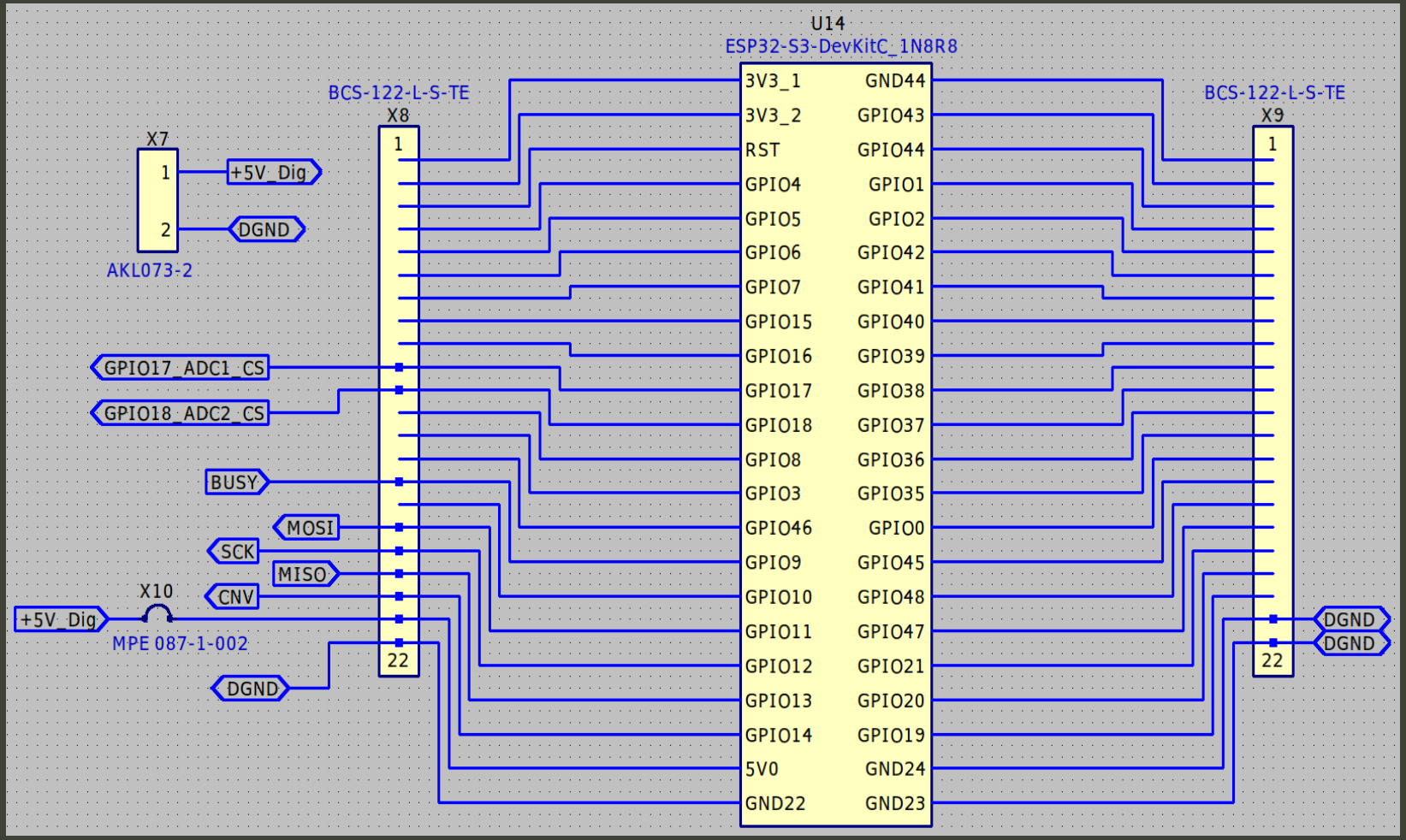
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- ***Fig. 5: ADC #2***



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- ***Fig. 6: Micro-Controller***

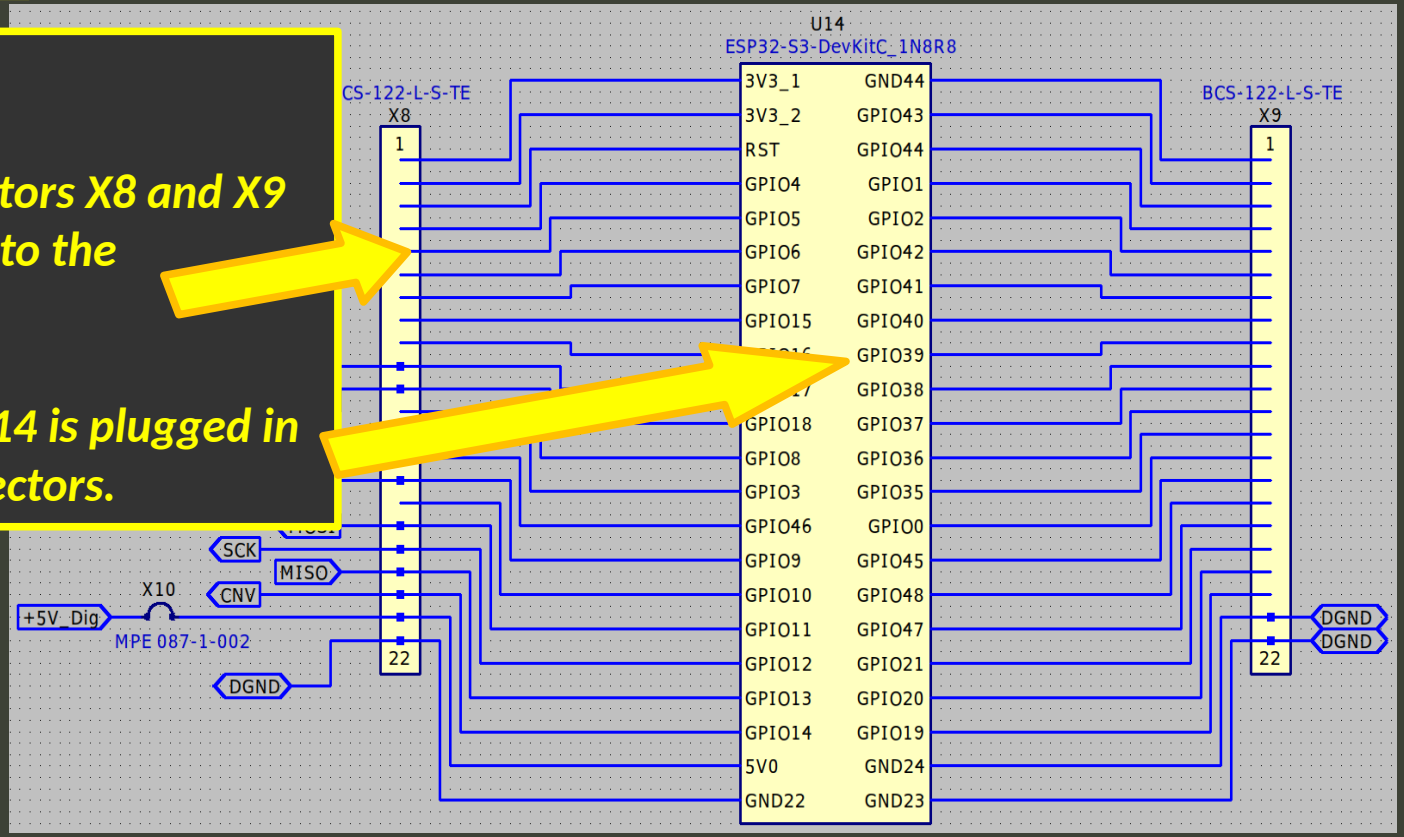


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- ***Fig. 7: Micro-Controller***

Important:

- ***Only the socket connectors X8 and X9 need to be transferred to the schematic in KiCad.***
- ***The micro-controller U14 is plugged in into these socket connectors.***

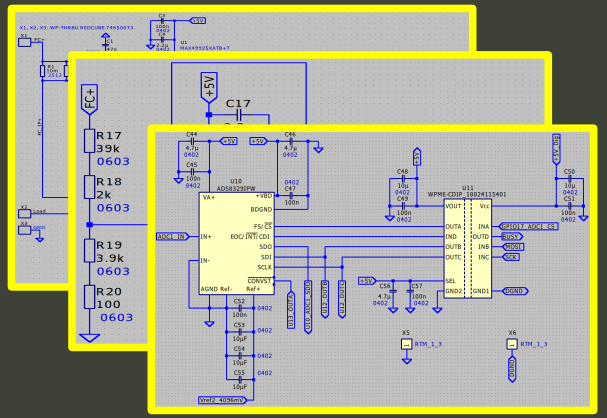


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- ***Analog circuits → exercises:***
  - Exercise 11: Figure 1 (see the previous slides) shows the fuel cell current measurement circuits. Calculate the voltage at the ADC1_IN measurement point in millivolts, when the load current of the fuel cell is 1A! Include your detailed solution (with step-by-step calculation) for this exercise in your project report!

→ *TASK: Solve exercise 11!*

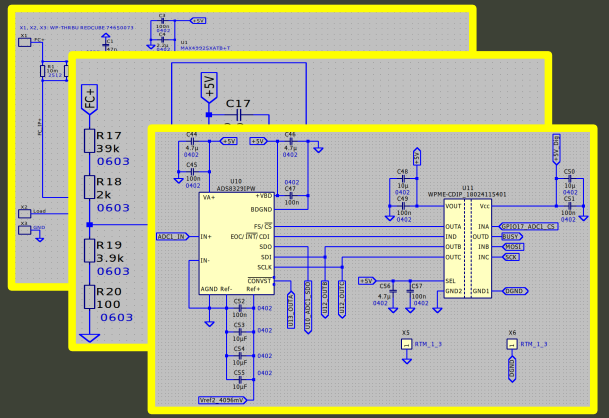


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- ***Analog circuits → exercises:***
  - Exercise 12: Figure 1 (see the previous slides) shows the fuel cell current measurement circuits. Calculate the voltage at the V_Isense (pin 1 of X4) measurement point in millivolts, when the load current of the fuel cell is 1A! Include your detailed solution (with step-by-step calculation) for this exercise in your project report!

→ *TASK: Solve exercise 12!*

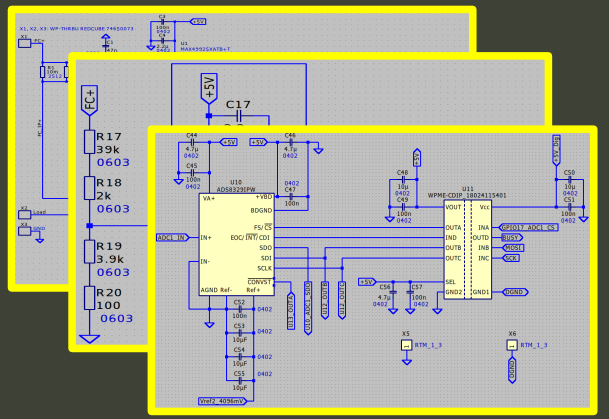


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- ***Analog circuits → exercises:***
  - Exercise 13: Figure 2 (see the previous slides) shows the fuel cell output voltage measurement circuits. Calculate the voltage at the ADC2_IN measurement point in millivolts, when the output voltage of the fuel cell is 30V! Include your detailed solution (with step-by-step calculation) for this exercise in your project report!

→ *TASK: Solve exercise 13!*



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- ***Lessons and tasks:***
  - independent study – repetition & solving exercises
  - until 09.11.2026: Symbols and footprints

    → In KiCad, create both symbol libraries and footprint libraries for all components listed in exercise 10. Important: Check the symbols and footprints multiple times!!!
  - until 09.11.2026: Schematic

    → In KiCad, create a project named “Fuel Cell Measurement Board” and within that project, create the schematic for the circuit board based on figures 1 through 6! The schematic must be created in DINA4-format and should be very easy to read when printed! Also very important: Check the schematic multiple times (Net-Classes, Net-Lables, ERC, etc.)!!!
  - Lesson 5 – Working on project

… schedule is continously updated!

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Thank you !
Have a nice day!



